

Structure-Adaptive Sparse Diffusion in Voxel Space for 3D Medical Image Enhancement

01 3D Medical Image Enhancement via Diffusion Models

Challenge: Efficiency vs Fidelity

- Latent / wavelet diffusion is cheaper but **loses fine anatomy** --> 3D patches in **voxel space**.
- Commonly-used noise/velocity targets are off data manifold, **amplified in 3D**, lead to slow, unstable training.

Challenge: Unconditional vs Conditional

- Enhancement is **conditional**. The degraded input already carries **strong anatomical priors**.
- With on-manifold x_0 prediction, adjacent steps in unconditional schedule ($T=1000$), **have been proved to be redundant**.

Challenge: Global vs Local

- Anatomy is heterogeneous**: homogeneous liver and intricate vessels have **different denoising dynamics** from patch to patch.
- One shared schedule can't fit every patch, so we need **structure-adaptive modulation**.

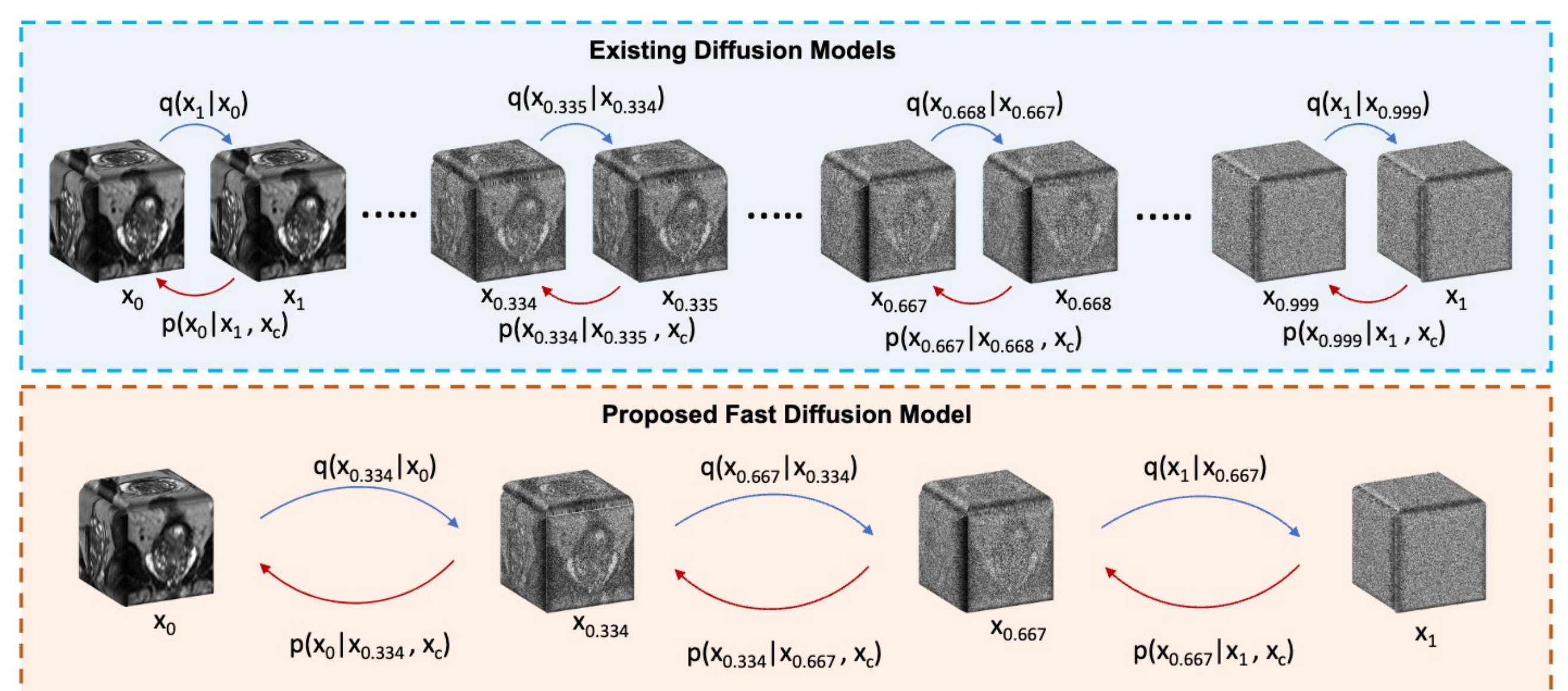
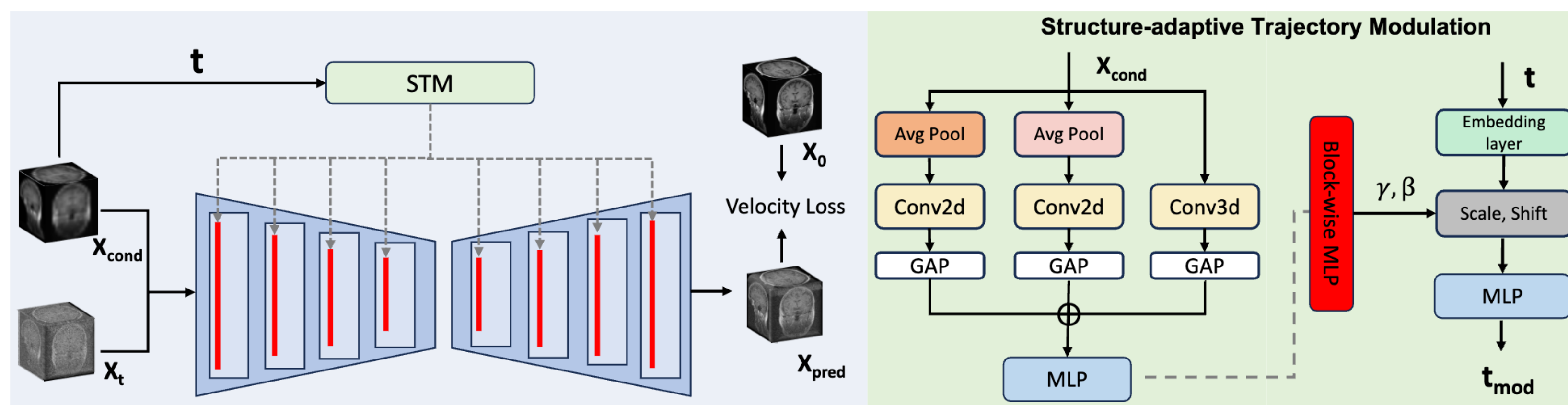
02 Methodology

① x_0 -Prediction + Velocity Supervision

- Predict clean x_0 **directly in voxel space**, so the **output stays on the data manifold**.
- Velocity loss $\|v_t - \hat{v}_t\|^2$ gives **uniformly scaled gradients**, offering **faster and more stable training**.

② Sparse Diffusion Time Scheduling

- $S = \{s_1, \dots, s_K\}$, $K \ll T = 1000$.
- The degraded input serves as a prior, so **a few steps suffice**. Train & sample only on S : **no redundant steps, faster training and model convergence**.



③ Structure-aware Trajectory Modulation (STM)

$$(\gamma_k, \beta_k) = \text{MLP}_k(c), t_{\text{mod}}^{(k)} = (1 + \gamma_k) \odot e_t + \beta_k$$

- A tri-planar encoder turns x_{cond} into a structure representation c , and each UNet block rescales its time embedding.
- The shared sparse schedule adapts to each patch's anatomy, **yielding per-patch denoising trajectories**.

03 Qualitative & Quantitative Results

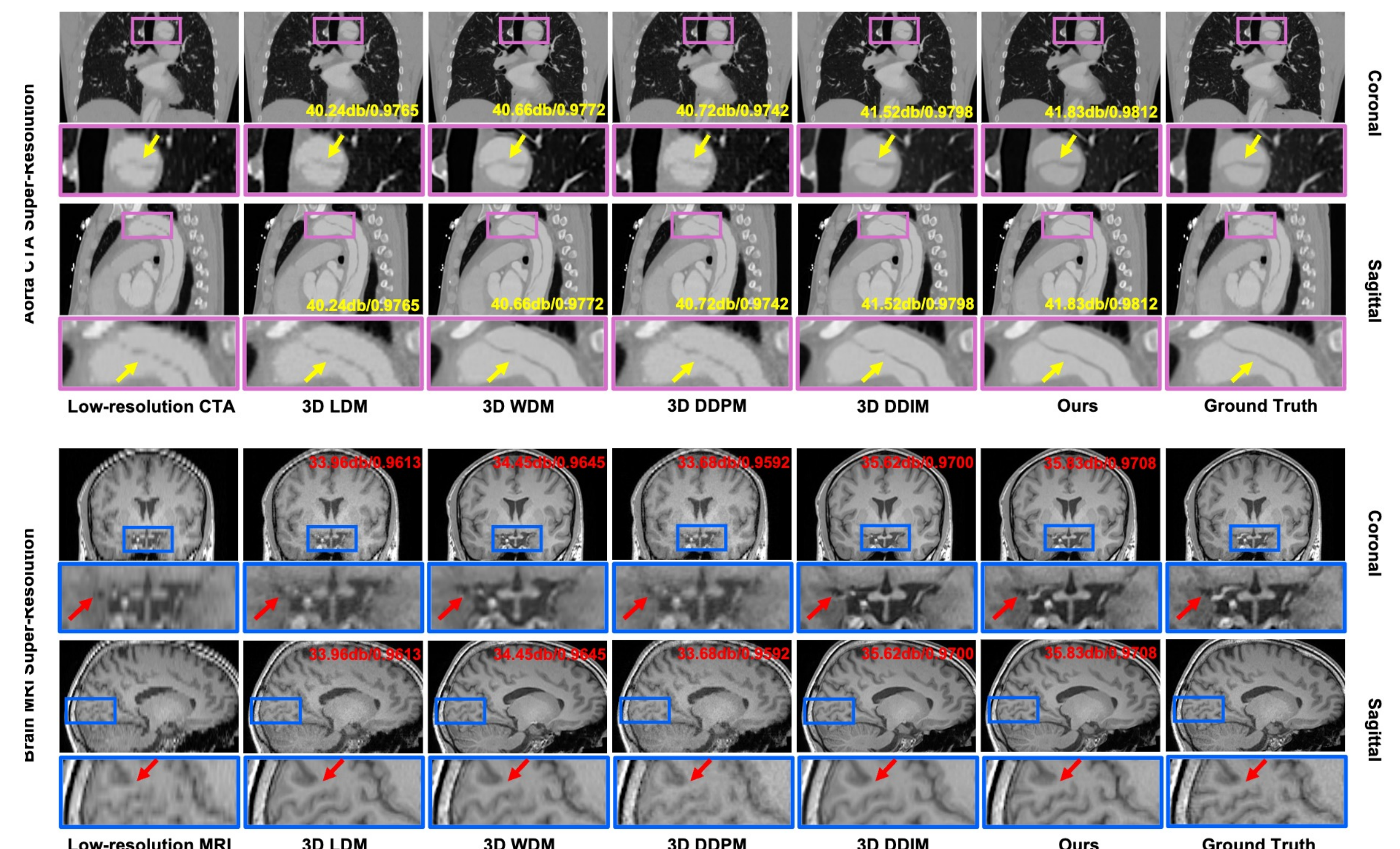
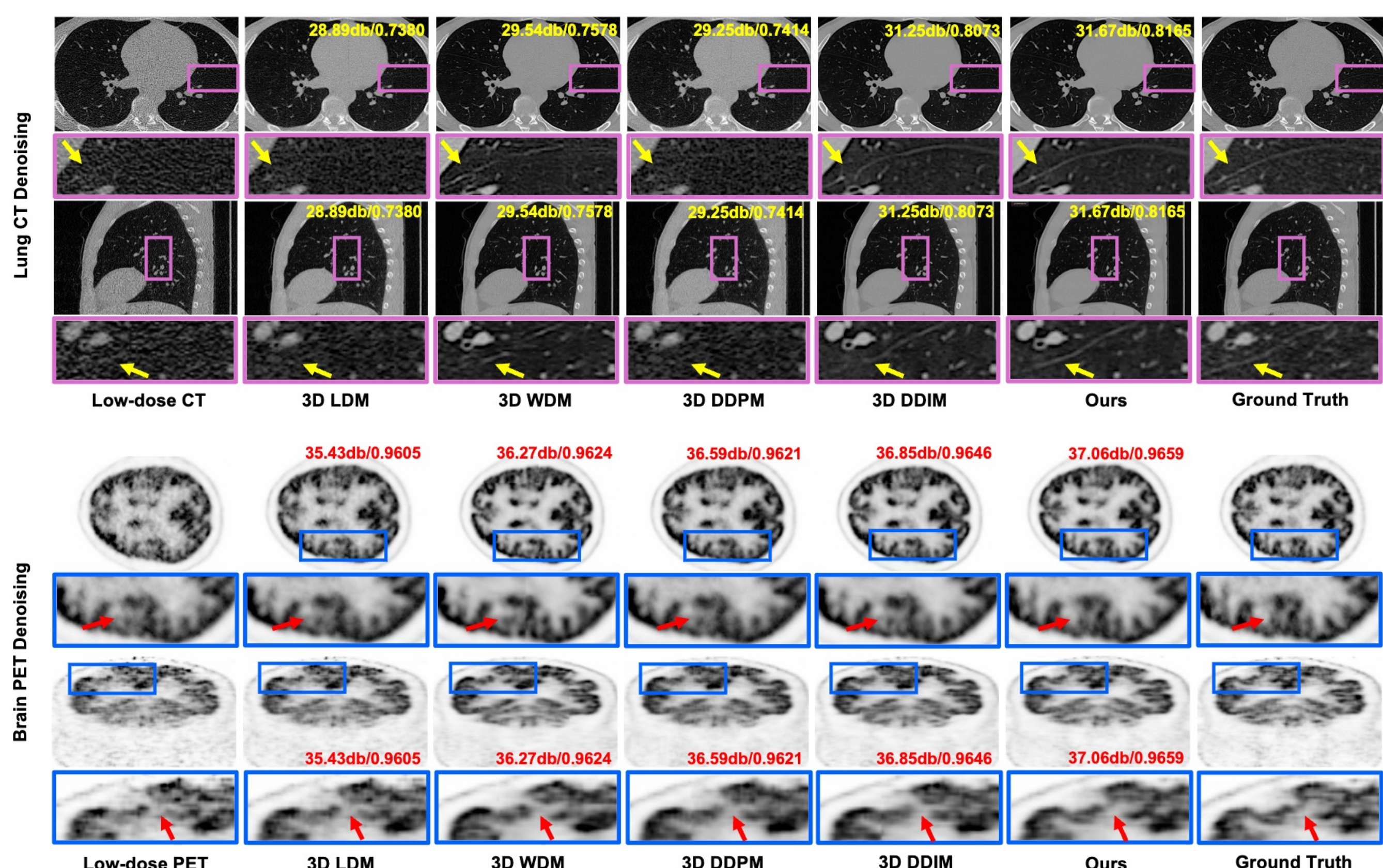
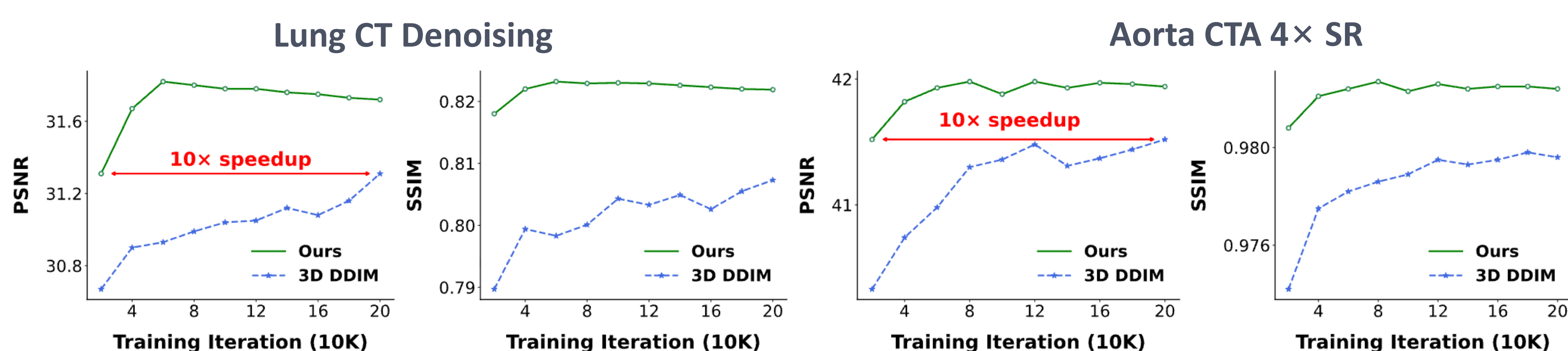


Table 1: best in **orange**, second in **blue**. All four metrics are **3D**. Train/test splits: Lung CT 40/10, Brain PET 70/18, Aorta CTA 50/10, Brain MRI 96/24.

Method	Lung CT Denoising				Brain PET Denoising				Aorta CTA 4× SR				Brain MRI 4× SR			
	PSNR↑	SSIM↑	MS-SSIM↑	HFEN↓	PSNR↑	SSIM↑	MS-SSIM↑	HFEN↓	PSNR↑	SSIM↑	MS-SSIM↑	HFEN↓	PSNR↑	SSIM↑	MS-SSIM↑	HFEN↓
3D LDM	28.89	0.7380	0.9430	0.0308	35.43	0.9605	0.9872	0.0179	40.24	0.9765	0.9951	0.0111	33.96	0.9613	0.9908	0.0207
3D WDM	29.54	0.7578	0.9532	0.0283	36.27	0.9624	0.9884	0.0167	40.66	0.9772	0.9952	0.0107	34.45	0.9645	0.9912	0.0200
3D DDPM	29.25	0.7414	0.9426	0.0291	36.59	0.9621	0.9890	0.0159	40.72	0.9742	0.9947	0.0107	33.68	0.9592	0.9903	0.0217
3D DDIM	31.25	0.8073	0.9613	0.0230	36.85	0.9646	0.9896	0.0156	41.52	0.9798	0.9957	0.0096	35.62	0.9700	0.9930	0.0172
Ours	31.82	0.8232	0.9644	0.0215	37.08	0.9660	0.9899	0.0149	41.98	0.9827	0.9965	0.0089	35.95	0.9724	0.9934	0.0163

04 Training Efficiency

Up to 10× faster training



x_0+v	Sparse	STM	PSNR↑	SSIM↑	Train time↓
×	×	×	19.42	.4639	1×
✓	×	×	31.25	.8073	1×
✓	✓	×	31.56	.8137	~0.1×
✓	✓	✓	31.82	.8232	~0.1×